

Consumer Credit Regulation and Lender Market Power

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The views expressed here do not reflect those of the Bank of Canada.

Interest Rate Ceilings

- The unsecured debt market is frequently the target of regulatory action (big credit card companies, payday lenders)
- Interest rate ceilings are a common regulatory tool, especially for high cost (payday) lenders
 - 43 states plus DC have explicit APR caps, 4 more have "unconscionability" restrictions (the caps depend on loan size, interstate branching restrictions are unconstitutional, and legal loopholes abound)
- There are large interest rate spreads: e.g. median payday loan APR is 322 percent
 - Default rate is high, but not nearly high enough to justify those rates
 - Market power? Operating costs?
- Caps are counterproductive in competitive lending markets with symmetric information

Question: What are the consequences of interest rate ceilings for consumer credit and welfare if lenders have **market power** and households may be **financially unsophisticated**?

This paper

- Develop a model of consumer credit and default that endogenously creates low and high-cost lenders
 - Partially directed search: informed and uninformed households
 - Informed: direct their search
 - Uninformed: randomly search
 - Encompasses different payday lending narratives: financial sophistication, market power, "pecking order" of credit
- Calibrate the model to match facts from the unsecured credit market, broadly defined (to include payday lending)
- Study the consequences of interest rate caps
 - We find that caps as low as 22 percent APR are welfare improving (optimal is around 25 percent, Pareto-improving)
 - There are spillovers into low-cost market for tight caps – rates rise in low-cost market as payday lenders exit, but negative effects are small

Environment

Main Ingredients

- Infinite horizon, small open endowment economy
- Unit measure of households
 - discount factor β , flow utility $u(c)$, idiosyncratic endowment process $y' = \rho_y y + \epsilon'$
 - borrow/save in one-period, non-contingent asset, a
 - default punished with utility penalty λ and one-period of financial autarky
- Information states $\{informed(I), uninformed(U)\}$
 - determines information in credit market
 - exogenous transition matrix π_{ij} for $i, j \in \{I, U\}$
 - not observable to lenders
- Large measure of risk-neutral intermediaries
 - can borrow/lend in world credit market at exogenous fixed interest rate r
 - to lend to households, must pay entry cost $\kappa > 0$

Credit Market with Partially Directed Search

Lester (2012) or Bethune, Choi, and Wright (2020)

- Intermediaries post schedules $\{a'_j, q_j\}_{j \in \{I, U\}}$ for household type (a, y) with commitment
 - (a, y) is observable and verifiable
 - $j \in \{I, U\}$ is unobservable
 - post direct revelation mechanisms
- Households take distribution of posted schedules as given
 - type I : direct their search to specific schedule
 - type U : search randomly
 - assume borrowers can always opt ex post to Kalai bargain (surplus share θ)
- Conjecture and verify equilibrium with at most 2 submarkets for each type (a, y)
 - “high-cost” market: cater to uninformed borrowers
 - “low-cost” market: compete for informed borrowers and screen for uninformed borrowers
 - pooling or separating possible

Household problem

- Default decision:

$$v(a, y, j) = \max \left\{ \underbrace{v^d(a, y, j)}_{\text{default}}, \underbrace{v^s(a, y, j)}_{\text{solvency}} \right\}$$

- Value of default

$$v^d(a, y, j) = u(y) - \lambda + \beta \mathbb{E}_{(y', j') | (y, j)} [v(0, y', j')]$$

- Value of saving

$$v^{s,n}(a, y, j) = \max_{c, a' \geq 0} \left\{ u(c) + \beta \mathbb{E}_{(y', j') | (y, j)} [v(a', y', j')] \right\}$$
$$c = a + y - \frac{1}{1+r} a'$$

- Value of borrowing in submarket $k = \{l, h\}$ at terms of trade (q_{jk}, a'_{jk})

$$v^{s,k}(a, y, j) = u(c) + \beta \mathbb{E}_{(y', j') | (y, j)} v(a'_{jk}, y', j')$$
$$c = a + y - q_{jk} a'_{jk}$$

Equilibrium in the Credit Market

Ex-post surpluses and tightness

- Lender surplus

$$\mathcal{S}^{\mathcal{L}}(q, a'; a, y, j) \equiv - \left[\frac{1 - \mathbb{E}_{(y', j')|(y, j)} [d(a', y', j')]}{1 + r} - q \right] a'$$

- Borrower surplus

$$\mathcal{S}^{\mathcal{B}}(q, a'; a, y, j) \equiv u(a + y - qa') + \beta \mathbb{E}_{(y', j')|(y, j)} [v(a', y', j')] - v^{s, n}(a, y, j)$$

- Tightness (lender-borrower ratio) in submarket (a, y, k) , where $k \in \{h, l\}$

$$n_k \equiv n(a, y, k)$$

Equilibrium in the Credit Market

High-Cost Submarket

- Terms of trade a'_{Uh}, q_{Uh} solve

$$\max_{a'_{Uh}, q_{Uh}} \left\{ \mathcal{S}^{\mathcal{L}}(q_{Uh}, a'_{Uh}; a, y, U) \right\}$$
$$S^{\mathcal{B}}(q_{Uh}, a'_{Uh}; a, y, U) \geq \bar{S}^{\mathcal{B}}(a, y, U)$$

- Free-entry condition pins down n_h :

$$\frac{\alpha(n_h)}{n_h} \mathcal{S}^{\mathcal{L}}(q_{Uh}, a'_{Uh}; a, y, U) = \kappa$$

- High-cost lenders extract the maximum possible surplus of uninformed borrowers
 - in equilibrium, only uninformed agents visit high-cost submarket
 - must verify informed would not be better off visiting the high-cost market

Equilibrium in the Credit Market

Bargaining Surpluses

- Let $\bar{S}^B(a, y, j)$ be the borrower's surplus that solves Kalai bargaining problem
- Maximizing borrower surplus

$$\max_{a', q} \left\{ S^B(q, a'; a, y, j) \right\}$$

- Borrower gets $\theta \in (0, 1]$ fraction of surplus:

$$(1 - \theta) S^B(q, a'; a, y, j) = \theta u'(a + y - qa') S^L(q, a'; a, y, j)$$

Equilibrium in the Credit Market

Low-Cost Submarket

- Terms of trade $(\{a'_{jl}, q_{jl}\}_{j \in \{I, U\}}, n_l)$ solve

$$\max_{\{a'_{jl}, q_{jl}\}_{j \in \{I, U\}}, n_l} \left\{ \alpha(n_l) \mathcal{S}^{\mathcal{B}}(q_{Il}, a'_{Il}; a, y, I) \right\}$$

$$\mathcal{S}^{\mathcal{B}}(q_{jl}, a'_{jl}; a, y, j) \geq \bar{\mathcal{S}}^{\mathcal{B}}(a, y, j) \quad \forall j \in \{I, U\}$$

$$\mathcal{S}^{\mathcal{B}}(q_{jl}, a'_{jl}; a, y, j) \geq \mathcal{S}^{\mathcal{B}}(q_{-jl}, a'_{-jl}; a, y, j) \quad \forall j$$

- Free entry condition:

$$\frac{\alpha(n_l)}{n_l} \sum_j \frac{\Gamma(a, y, j)}{\Gamma(a, y, I) + \Gamma(a, y, U)} \mathcal{S}^{\mathcal{L}}(q_{jl}, a'_{jl}; a, y, j) = \kappa$$

- Low-cost lenders compete for informed agents
 - competition may be limited by desire to screen uninformed
 - we can get separating and pooling outcomes in equilibrium

Equilibrium in the Credit Market

Probability of matching

- Value of solvency

$$v^s(a, y, j) = \underbrace{\alpha_l v^{s,l}(a, y, j)}_{\text{match low-cost}} + \underbrace{\alpha_h v^{s,h}(a, y, j)}_{\text{match high-cost}} + \underbrace{(1 - \alpha_l - \alpha_h) v^{s,n}(a, y, j)}_{\text{unmatched, can save } a' \geq 0}$$

- Probability that household (a, y, j) matches in submarket $i \in \{l, h\}$

$$\alpha_l = \begin{cases} \alpha(n_l) & \text{if } j = I \\ \frac{N_l}{N_l + N_h} \alpha(n_l) & \text{if } j = U \end{cases} \quad \alpha_h = \begin{cases} 0 & \text{if } j = I \\ \frac{N_h}{N_l + N_h} \alpha(n_h) & \text{if } j = U \end{cases}$$

- Households ex ante credit schedule depends on the composition of high- and low-cost lenders

Equilibrium properties in credit market

Lemma

If the high-cost submarket is active, $N_h(a, y) > 0$, for any $(a, y) \in \text{supp}\Gamma(a, y)$, then

① $S^{\mathcal{L}}(q_{jl}, a'_{jl}; a, y, j) < S^{\mathcal{L}}(q_{Uh}, a'_{Uh}; a, y, U)$ for $j \in \{I, U\}$

② $n_l < n_h$

- high-cost lenders must make higher surplus conditional on trade
- implies they have lower expected prob of trade
- numerically, we find $q_{Il} \leq q_{Ul} < q_{Uh}$ and $a'_{Il} \leq a'_{Ul} < a'_{Uh}$

Calibration

Strategy and Results

- We match features of the broad unsecured credit market, including high-cost lending
 - **standard targets:** fraction of < 0 net worth, average default rate, average debt-to-income
 - **high-cost targets:** % borrowing high cost, interest rate spread, length of high-cost sequences
- Jointly determine:
 - discount factor β
 - disutility from default λ
 - entry cost κ
 - curvature of matching function v
 - information shock process π_{II}, π_{UU}

Calibration

Strategy & Results

We match features of the broad unsecured credit market, including high-cost lending

- **standard targets:** fraction of < 0 net worth, average default rate, average debt-to-income
- **new targets:** % borrowing high cost, interest rate spread, length of high-cost sequences

Table: Targeted moments: data and model

Moment	Data	Model	Parameter	Value
Fraction $NW < 0$ (%)	12.50	14.44	entry cost, κ	7e-5
Default rate (annualized, %)	1.20	1.30	default cost, λ	0.012
Debt-to-income ratio (%)	1.23	1.31	discount, β	0.991
Mean Rate Low-cost (annualized, %)	12.07	10.98	matching curvature, v	28.39
Mean Rate High-cost (annualized, %)	339	304	bargaining, θ	0.01
Fraction High-cost (prev. 12m, %)	3.4	3.8	informed per., π_{II}	0.97
Fraction of High-cost Borrowing Sequences	0.55	0.67	uninformed per., π_{UU}	0.41

Distribution of interest rates

- Interest rates do not overlap across submarkets; highest rate in low-cost market is 60 percent

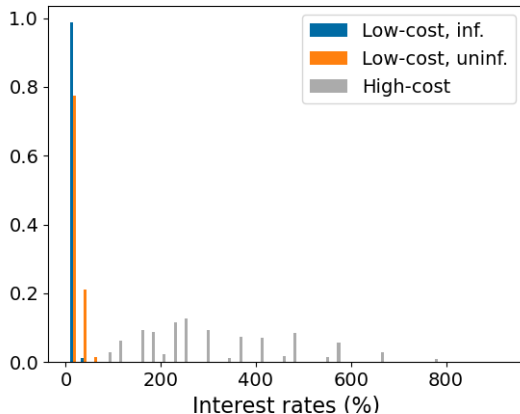
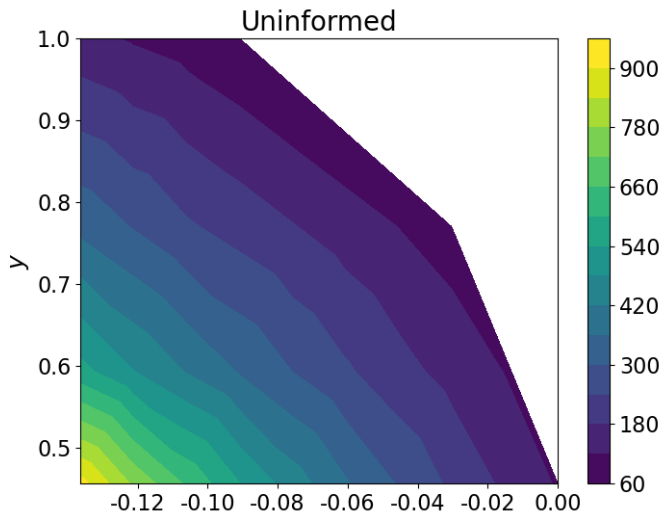


Figure: Relative frequency of interest rates, by submarket

Interest rates in high-cost markets

- Low-wealth, low-income households face higher interest rates



Endogenous market segmentation

- Two submarkets—high and low-cost—are active for low wealth
- High-cost submarket is more prevalent as wealth falls (consistent with data from the SCF)

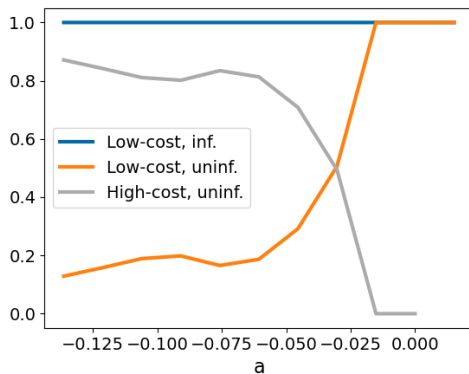


Figure: Probability of searching and matching in a submarket (fixed y)

The Composition of Interest Rates

- Compare risk-based (justified by default rates), +fixed costs, and market power

	Low-cost, inf	Low-cost, un	High-cost, un
Effective	10.8	26.0	303.9
Risk-based	10.1	5.5	8.3
Risk-based+Fixed Cost	11.4	12.1	9.6

- Within low-cost market, informed agents are subsidized; uninformed borrow less and default less, but pay higher share of fixed cost and market share premia

Interest Rate Caps

- Interest rate caps are a common regulatory tool
- In perfectly-competitive default models, caps shrink budget sets and therefore are weakly welfare-reducing
- In pure competitive search models, caps simply add a constraint to the lender problem, weakly welfare-reducing (competitive search already maximizes borrower welfare subject to free entry)
- An unrestricted planner would set the cap to the competitive search outcome for each borrower
- In our model, planner cannot regulate "by lender type" easily, so blunt interest rate caps are a natural tool

Aggregate welfare and interest rate ceilings

- Welfare gains from ceilings as low as 30 percent APR
- High-cost market shuts down for ceilings below 22 percent APR

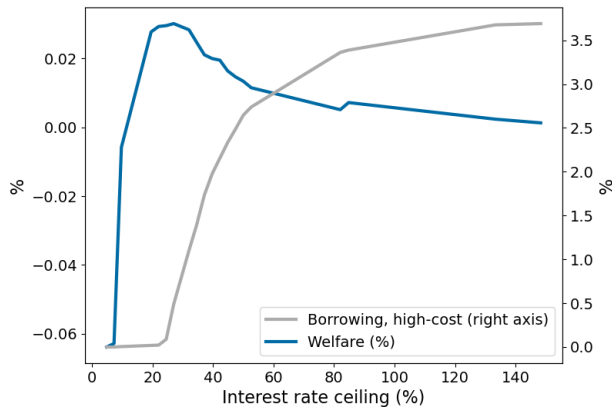


Figure: Debt, by submarket

Effects of ceilings in high-cost market

- Ceilings are binding for the high-cost submarket, which reduces lender entry

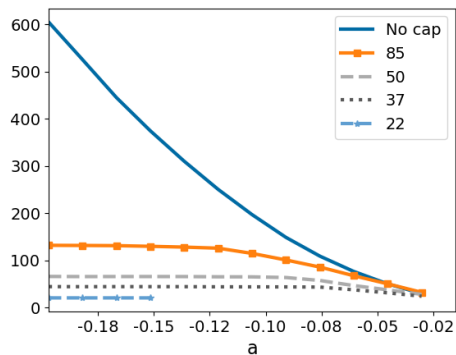
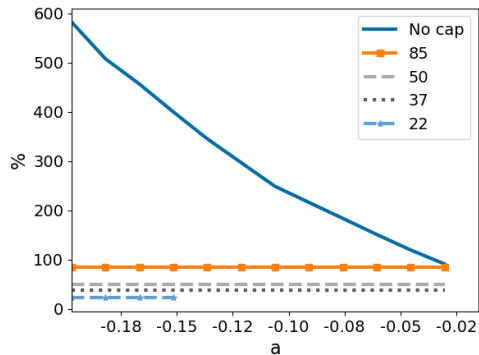


Figure: Price and tightness, high-cost market, per interest rate cap (fixed y)

Spillover effects on market composition

- Ceilings tend to spill over to the low-cost submarket, increasing the probability of matching in low-cost submarkets when uninformed

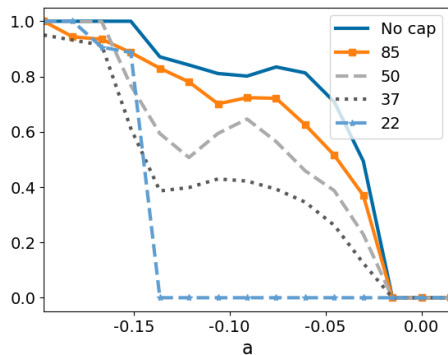
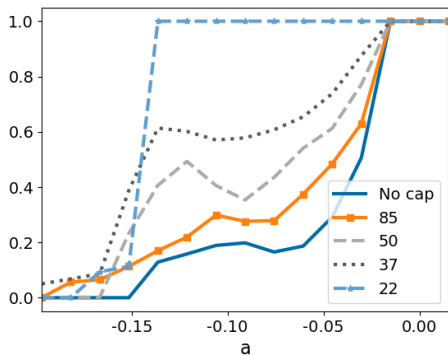


Figure: Ex-ante probability of matching, per interest rate cap and submarket (fixed y)

Spillover effects on borrowing rates

- More households borrow in low-cost submarkets—both informed and uninformed; borrowing in high-cost markets is reduced and eventually shuts down

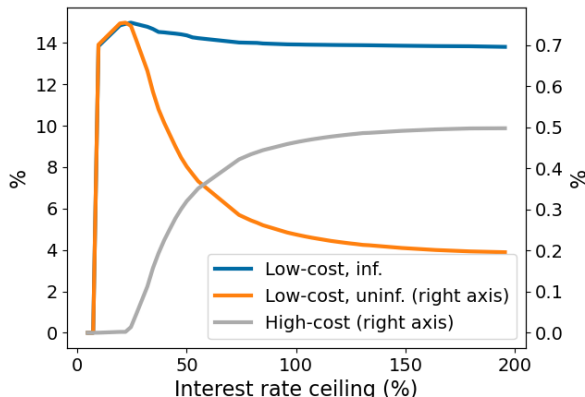


Figure: Fraction of households borrowing, by submarket

Spillover effects on interest rates

- Higher interest rates in low-cost markets despite the interest rate caps

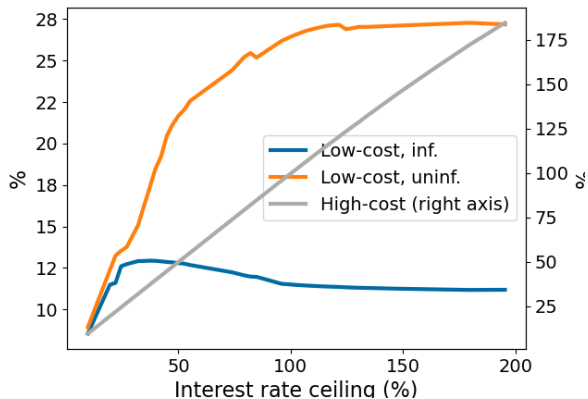


Figure: Interest Rates, by submarket (fixed y)

Interest rates rise (in low-cost) primarily due to default risk

- Higher rates in low-cost markets explained by higher default rates due to more borrowing (intensive margin)

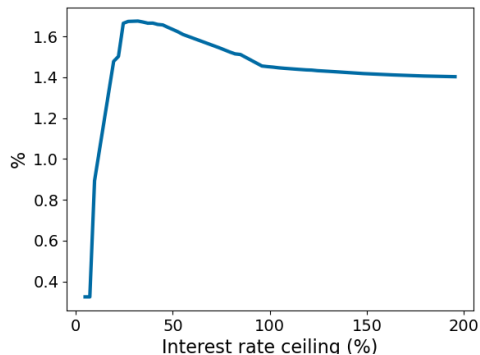
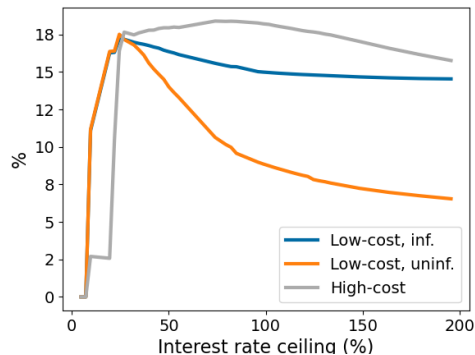


Figure: Debt-to-income and default rate

Welfare effects in the cross-section

- We find Pareto improvements with ceilings, largest gain by lowest (a, y) uninformed

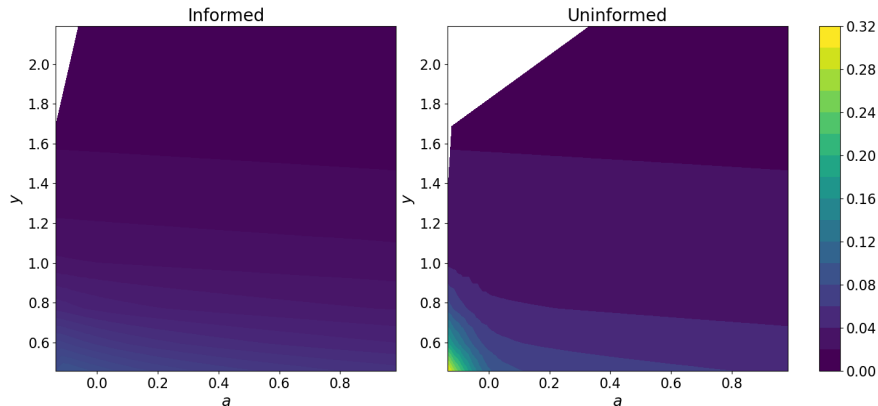


Figure: Welfare gain of 22 percent ceiling

Spillover effects in the cross-section

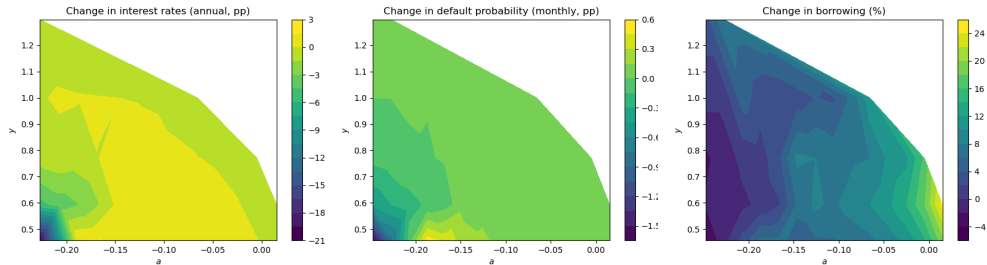


Figure: Change in APRs (left), default prob. (middle), and borrowing (right), from a 64 percent ceiling

Interest Rate Ceilings

Recap

- As cap gets tighter, profit conditional on trade falls in high cost market
- Lenders exit high-cost market, more uninformed borrowers end up in low-cost market (better market composition)
- More households borrow—intensive and extensive margin—from the low-cost market and less in the high-cost market, improving consumption smoothing
- Interest rate ceilings can (Pareto) improve welfare

How good are caps? Welfare gain from eliminating uninformed agents

- Caps may be better than nothing, but can we do better?
- Not much better – competitive search gain 0.032 percent, optimal ceiling 0.03 percent

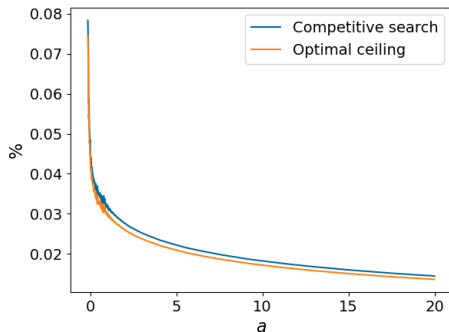


Figure: Welfare gain of competitive search allocation

Conclusion

- We embed a partially-directed search model in an Eaton-Gersovitz model to study market power in the unsecured credit market
- We calibrate the model to match the key features of the unsecured credit market (broadly defined to include payday lenders)
- We find that interest rate caps can Pareto improve welfare and capture nearly all the gains available (almost get us the competitive search level of welfare on average)